

Example 14: Let $Q(x)$ denote the statement " $x = x + 1$." What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?

Soln: $x = x + 1 \Rightarrow 1 = 0$, Which is False. $\therefore \exists x Q(x)$ is False.

Remark: Generally, an implicit assumption is made that all domains of discourse for quantifiers are nonempty. If the domain is empty, then $\exists x Q(x)$ is false whenever $Q(x)$ is a propositional function because when the domain is empty, there can be no element x in the domain for which $Q(x)$ is true.

Imp.

The Uniqueness Quantifier

The notation $\exists! x P(x)$ [or $\exists x P(x)$] states that "There exists a unique x such that $P(x)$ is true." For instance, $\exists! x (x - 1 = 0)$, where the domain is the set of real numbers, states that there is a unique real number x s.t. $(x - 1) = 0$. This is a true statement, $\because x = 1$ is that unique real number.

1.4.4 Quantifiers Over Finite Domains

When the domain of a quantifier is finite, that is, when all its elements can be listed, quantified statements can be expressed using propositional logic. In particular, when the elements of the domain are $x_1, x_2, \dots, x_n$, where n is a positive integer, the universal quantification $\forall x P(x)$ is the same as the conjunction $[P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)]$, because this conjunction is true if and only if $P(x_1), P(x_2), \dots, P(x_n)$ are all true.

Example 15: What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the domain consists of the positive integers not exceeding 4?

Soln: $\forall x P(x)$ is equivalent to $P(1) \wedge P(2) \wedge P(3) \wedge P(4) \dots$
 $P(1) \because 1^2 = 1 < 10 \rightarrow \text{True}$ $P(2) \because 2^2 = 4 < 10 \rightarrow \text{True}$
 $P(3) \because 3^2 = 9 < 10 \rightarrow \text{True}$ $P(4) \because 4^2 = 16 < 10 \rightarrow \text{False}$
 $\therefore \forall x P(x)$ is False.